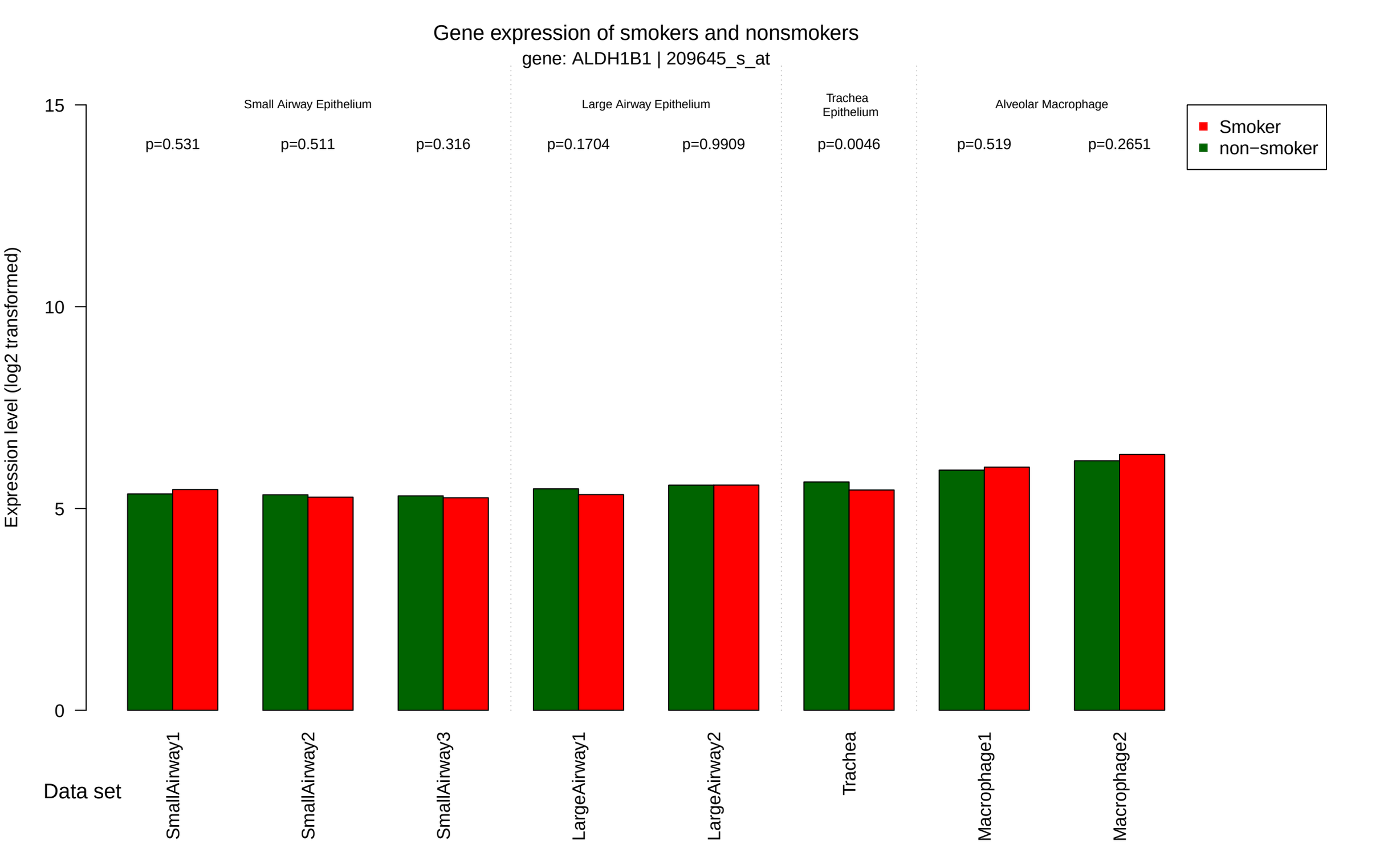
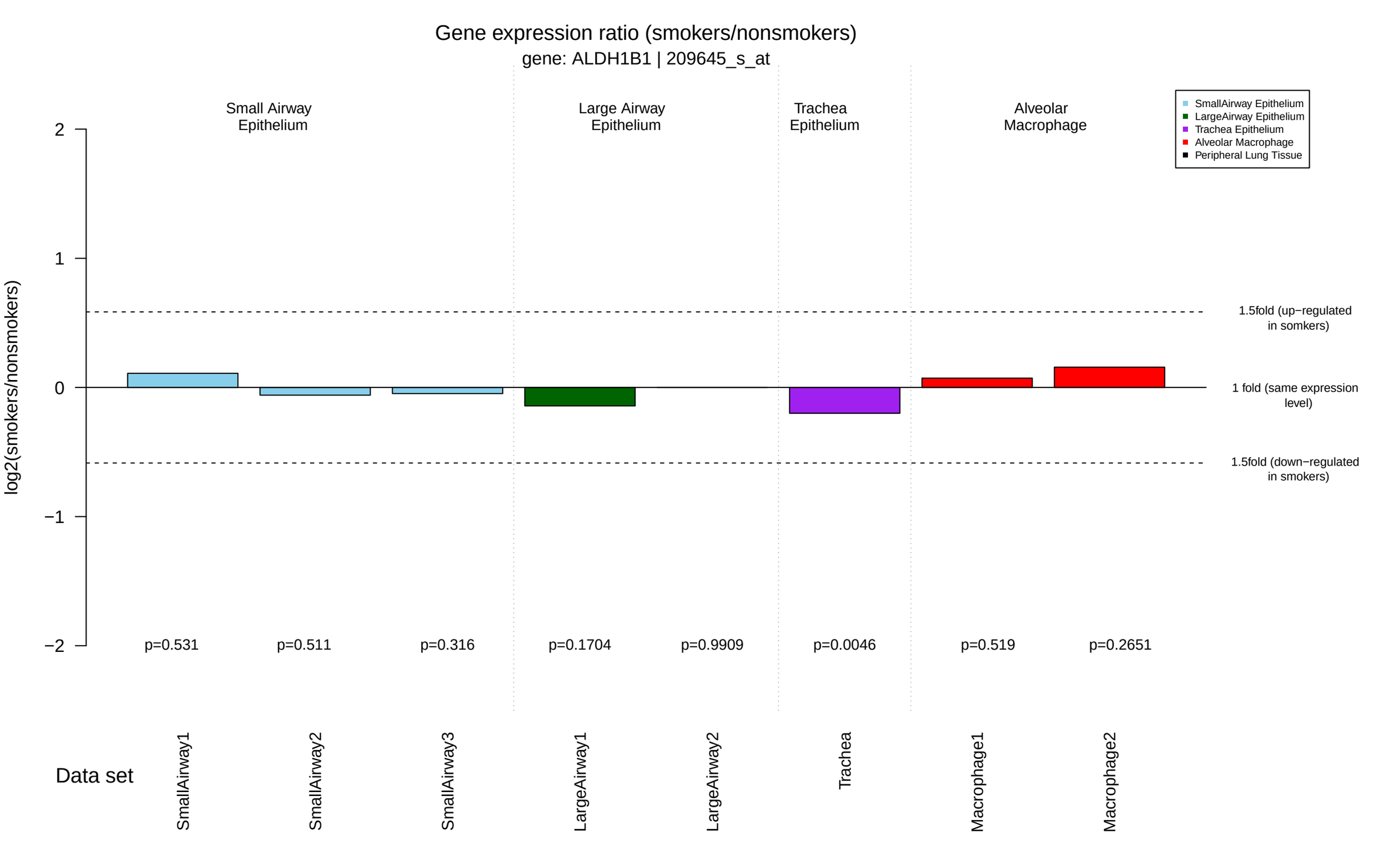
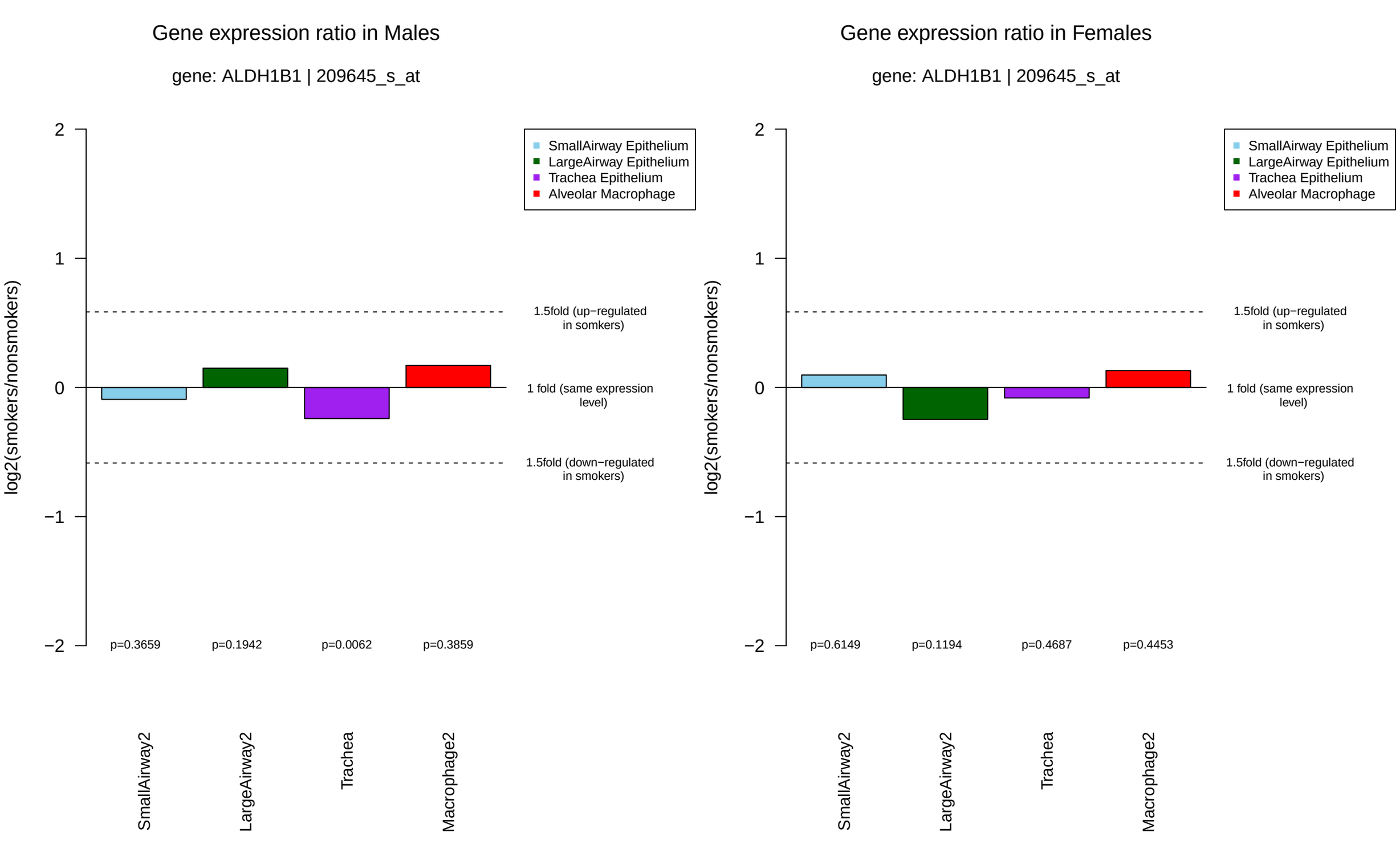
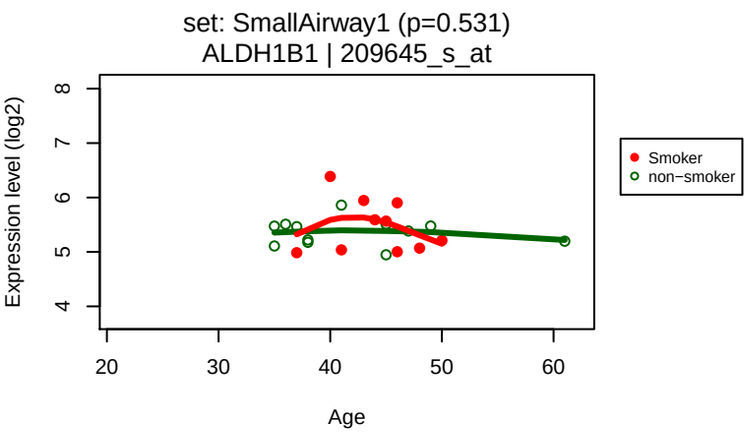
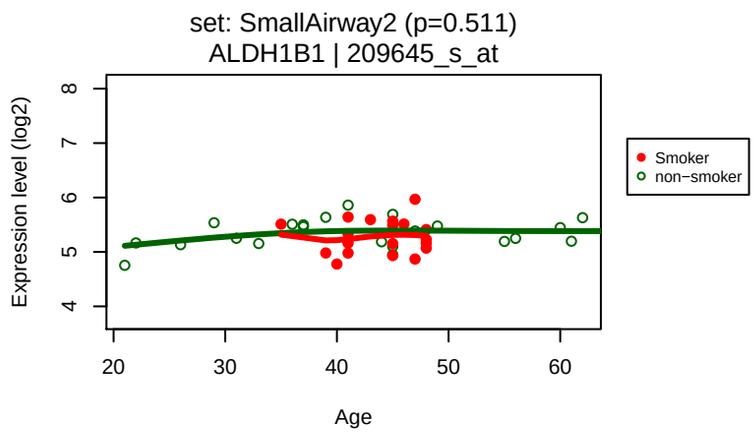
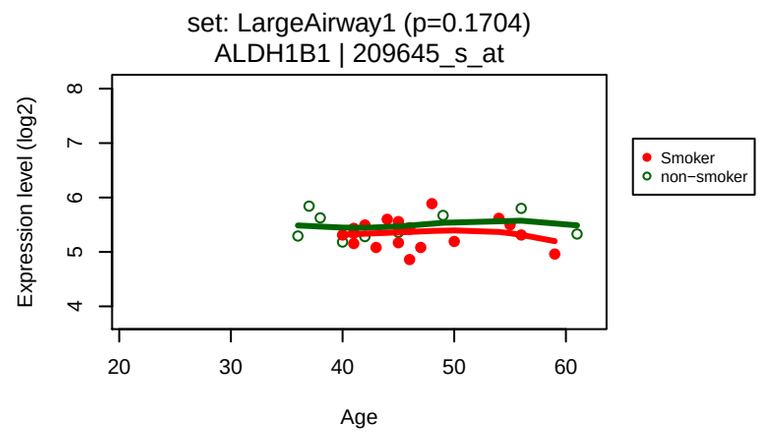

Gene expression of smokers and nonsmokers
gene: ALDH1B1 | 209645_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.531
p=0.511
p=0.316
p=0.1704
p=0.9909
p=0.0046
p=0.519
p=0.2651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

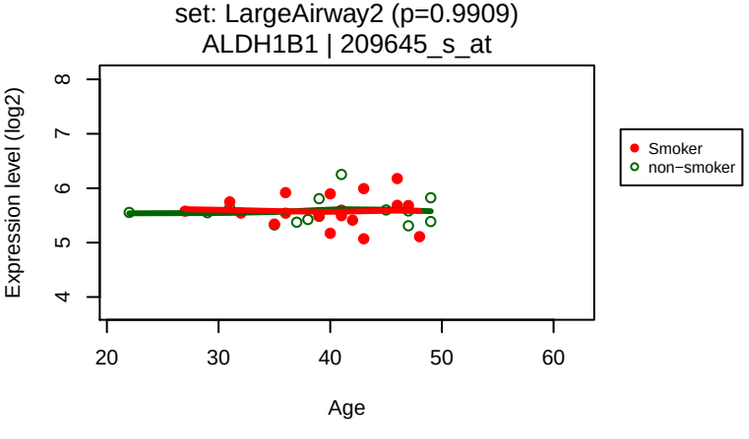

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1B1 | 209645_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.531
p=0.511
p=0.316
p=0.1704
p=0.9909
p=0.0046
p=0.519
p=0.2651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

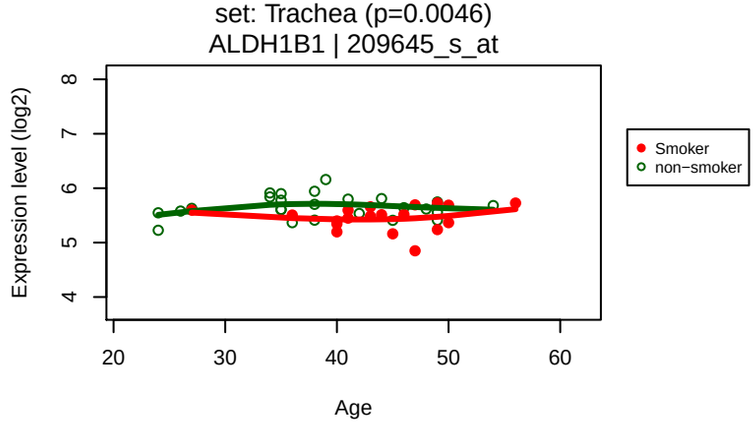

Gene expression ratio in Males
gene: ALDH1B1 | 209645_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3659
p=0.1942
p=0.0062
p=0.3859
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1B1 | 209645_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6149
p=0.1194
p=0.4687
p=0.4453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

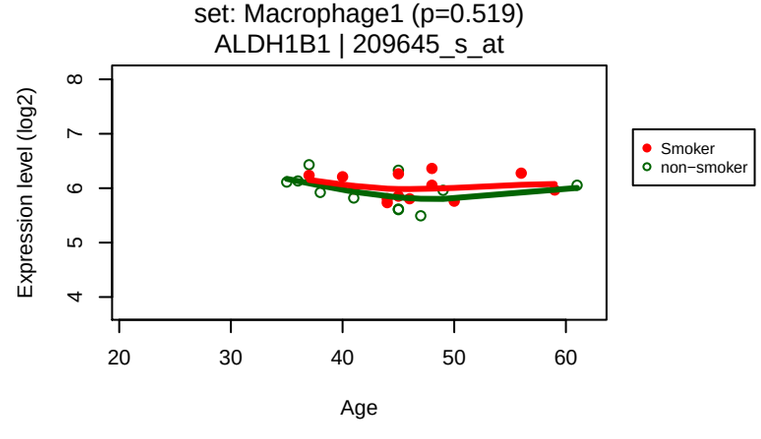

set: SmallAirway1 (p=0.531)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker

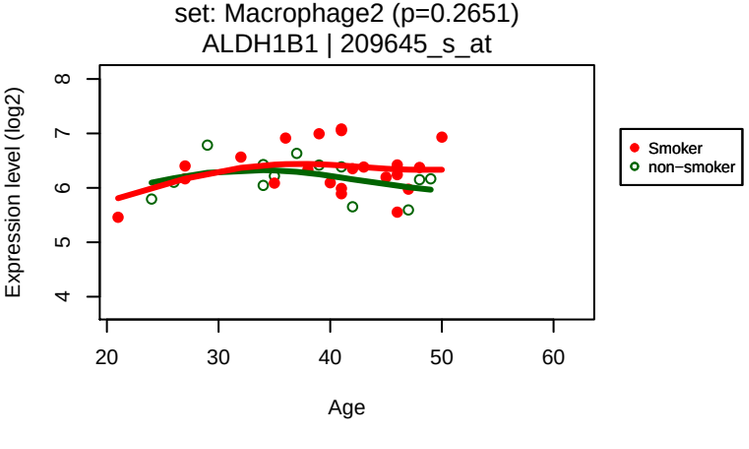

set: SmallAirway2 (p=0.511)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker

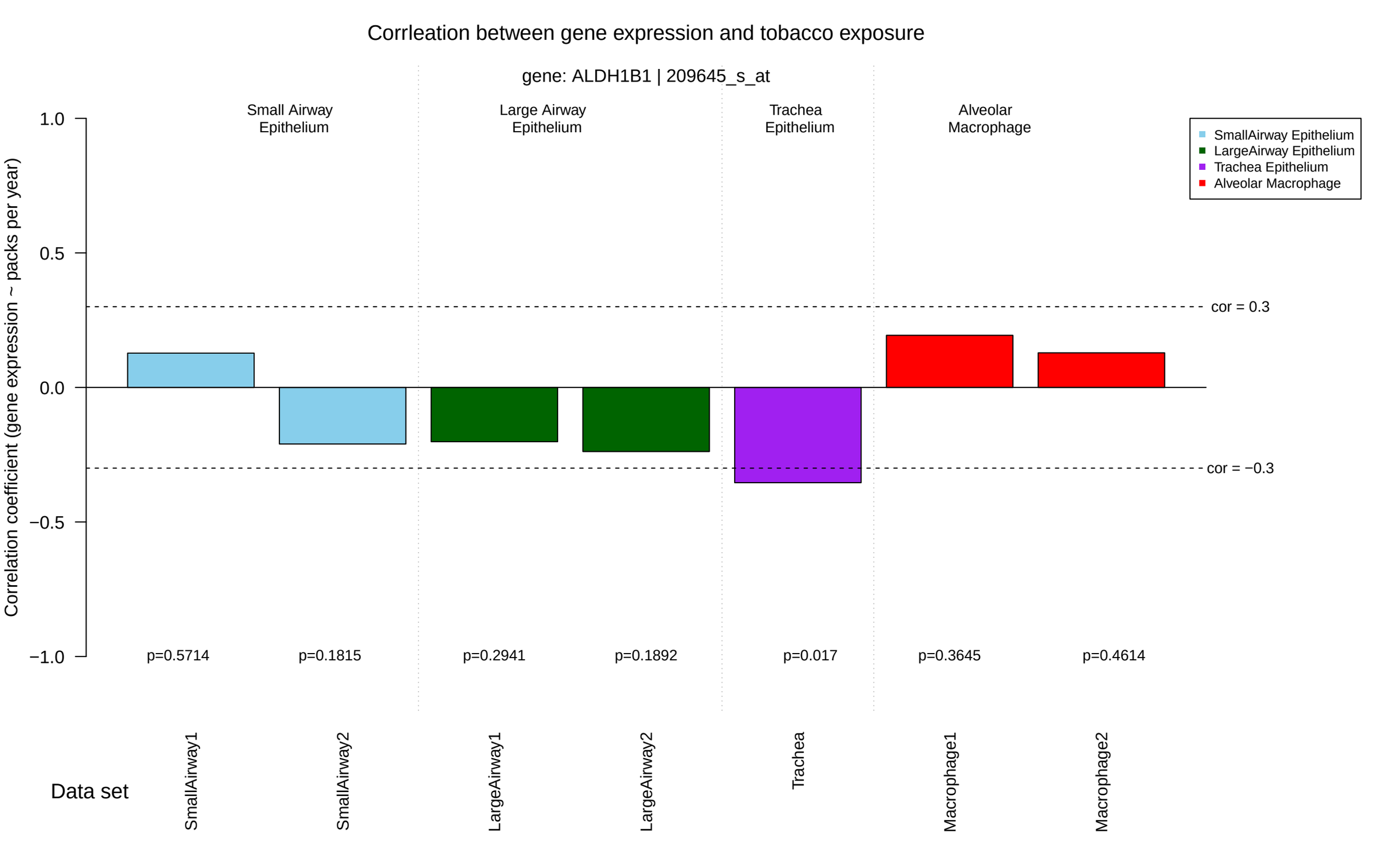

set: LargeAirway1 (p=0.1704)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9909)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.519)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2651)
ALDH1B1 | 209645_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1B1 | 209645_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5714
p=0.1815
p=0.2941
p=0.1892
p=0.017
p=0.3645
p=0.4614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2